JEE Mathematics: Quick Review with Grading and Practice Map - Matrices (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (111 questions; a question that appears in two of these is counted once): High = 10 or more questions, Medium = 1 to 9 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Definition of Matrix

Question 01

Are you able to define the order of a matrix and how its elements are represented?

Model Answer

A matrix is a rectangular array of numbers in m rows and n columns; its order is $m \times n$ and it has mn elements. a_{ij} is the element in row i, column j, and $A = [a_{ij}]_{m\times n}$.

Ex: 3×2 matrix with $a_{ij} = \frac{1}{2}|i - 3j|$ is $\begin{pmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{pmatrix}$.

Grade - Medium

Practice map, main concept: Sheet: J-A 26; C-I 11, 13

Also used in: Sheet: J-II 1

Question 02

Can you find the possible number of orders of a matrix if its number of elements is known?

Model Answer

If there are N elements, every ordered factor pair $N = m \times n$ is a possible order, so the number of orders = the number of positive divisors of N (a prime N gives only $1 \times N$ and $N \times 1$).

Ex: 12 elements $\Rightarrow$ 6 orders: $1 \times 12, 2 \times 6, 3 \times 4, 4 \times 3, 6 \times 2, 12 \times 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Important Definitions

Question 03

Can you identify a row matrix and a column matrix?

Model Answer

A row matrix has one row (order $1 \times n$); a column matrix has one column (order $m \times 1$).

Ex: $\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$ is 1×3; $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$ is 2×1. A column matrix X gives X^TX (a number) and XX^T (a square matrix).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04

Are you able to define null, horizontal, vertical and square matrices?

Model Answer

Null (zero) matrix O: every element is 0. Horizontal: $m < n$ (more columns). Vertical: $m > n$ (more rows). Square: $m = n$.

Ex: 2×3 is horizontal, 3×2 is vertical, 3×3 is square.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 05

What do principal diagonal, conjugate elements and trace of a square matrix mean?

Model Answer

Principal diagonal: $a_{11}, a_{22}, \ldots, a_{nn}$. Conjugate elements: the pair a_{ij}, a_{ji}. Trace: $\operatorname{tr} A = \sum a_{ii}$ (sum of the diagonal elements).

Ex: $\operatorname{tr}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 5$; $\operatorname{tr}(AB) = \operatorname{tr}(BA)$; $\operatorname{tr}(AA^T) = \sum a_{ij}^2$.

Grade - Medium

Practice map, main concept: DPP: 1(11)

Also used in: Sheet: J-II 22, 28-30; J-M 13, 17, 36, 39-40; J-A 18, 21-22; C-I 4; C-II 24 | Question Bank: 145, 161

Question 06

How do you check whether a matrix is singular (non-invertible) or non-singular (invertible)?

Model Answer

Only a square matrix can be tested: $|A| = 0 \Leftrightarrow$ singular (no inverse); $|A| \neq 0 \Leftrightarrow$ non-singular (inverse exists).

Ex: $\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$, so $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is singular.

Grade - Medium

Practice map, main concept: Sheet: J-II 28; J-M 2, 9; J-A 6; C-I 9 | DPP: 1(2) | Question Bank: 117, 135, 141

Also used in: Sheet: J-M 25; J-A 24, 26-27; C-I 19; C-II 8, 21, 24-25 | DPP: 4(8) | Question Bank: 123, 149

Special Types of Square Matrices

Question 07

How do you check whether a square matrix is diagonal, scalar, identity, upper triangular or lower triangular?

Model Answer

Diagonal: $a_{ij} = 0$ for $i \neq j$. Scalar: diagonal with all diagonal elements equal (kI). Identity I: scalar with $k = 1$. Upper triangular: $a_{ij} = 0$ for $i > j$. Lower triangular: $a_{ij} = 0$ for $i < j$.

Ex: for a triangular matrix of order n the maximum number of distinct entries is $\frac{n(n+1)}{2}$; for a scalar matrix $|kI_n| = k^n$.

Grade - Low

Practice map, main concept: Sheet: J-II 1 | DPP: 1(1, 7-9); 5(3)

Also used in: Sheet: J-II 14; C-I 17

Question 08
Do you know an alternate way to denote a diagonal matrix?

Model Answer
$\text{diag}(d_1, d_2, \ldots, d_n)$ lists the diagonal entries. Then $D^m = \text{diag}(d_1^m, \ldots, d_n^m)$, $D^{-1} = \text{diag}(1/d_1, \ldots, 1/d_n)$ and $|D| = d_1 d_2 \cdots d_n$.

Ex: $\text{diag}(2, 3)^3 = \text{diag}(8, 27)$.

Grade - Medium

Practice map, main concept: DPP: 5(4, 9)

Also used in: Sheet: J-A 1; C-I 17 | DPP: 4(10)

Equality of two Matrices

Question 09
What is the condition for two matrices to be equal?

Model Answer
$A = B$ iff they have the same order and $a_{ij} = b_{ij}$ for every i, j; equate corresponding elements to get equations.

Ex: $\begin{pmatrix} x+y & 2 \\ 3 & x-y \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix} \Rightarrow x = 3,\ y = 2.$

Grade - Low

Practice map, main concept: Sheet: J-II 3; C-I 8, 16 | DPP: 1(5, 10)

Also used in: Sheet: J-I 6; C-II 15

Addition/Subtraction of Matrices

Question 10
When can we add or subtract two matrices and how?

Model Answer
Only when they have the same order: $A \pm B = [a_{ij} \pm b_{ij}]$. Addition is commutative and associative: $A + B = B + A$, $(A + B) + C = A + (B + C)$.

Ex: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 4 & 4 \end{pmatrix}$; a 2×2 and a 2×3 matrix cannot be added.

Grade - Low

Practice map, main concept: DPP: 1(6, 12); 2(5)

Also used in: Sheet: J-II 3 | DPP: 1(3, 5, 13, 15); 2(15)

Question 11
What happens to the determinant value when we add two matrices?

Model Answer
Nothing simple: in general $|A + B| \neq |A| + |B|$; always add the matrices first, then take the determinant.

Ex: $A = B = I_2$: $|A + B| = |2I| = 4$ but $|A| + |B| = 2$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 12
What do you mean by the additive inverse in matrix algebra?

Model Answer
The additive inverse of A is $-A = [-a_{ij}]$, with $A + (-A) = O$; the zero matrix O is the additive identity, $A + O = A$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 13
What happens to the trace when we add two matrices?

Model Answer

$$\operatorname{tr}(A \pm B) = \operatorname{tr} A \pm \operatorname{tr} B, \qquad \operatorname{tr}(kA) = k \operatorname{tr} A$$

Ex: if $A + 2B$ and $2A - B$ have traces -1 and 3, then $\operatorname{tr} A = 1,\ \operatorname{tr} B = -1$ and $\operatorname{tr} A - \operatorname{tr} B = 2$.

Grade - Low
Practice map, main concept: Sheet: J-I 1; J-II 30
Also used in: Sheet: J-II 28

Multiplication of a Matrix by a scalar

Question 14
What happens when we multiply a matrix by a real number k?

Model Answer
Every element is multiplied by k: $kA = [ka_{ij}]$; $k(A \pm B) = kA \pm kB$ and $(k_1 + k_2)A = k_1A + k_2A$.
Ex: $3\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 6 & -3 \end{pmatrix}$.

Grade - Low
Practice map, main concept: DPP: 1(3-4, 13-15)
Also used in: Sheet: J-II 3 | DPP: 1(5)

Question 15
What happens to the determinant value when we multiply a matrix by a real number k?

Model Answer

$$|kA| = k^n|A| \quad (A \text{ of order } n \times n)$$

Ex: for 3×3 A with $|A| = 2$: $|2A| = 2^3 \cdot 2 = 16$, $|-A| = -2$.

Grade - Medium
Practice map, main concept: Sheet: J-M 34
Also used in: Sheet: J-I 24; J-M 22 | DPP: 5(3)

Multiplication of Matrices

Question 16
What do pre-factor and post-factor mean in matrix multiplication?

Model Answer
In AB, A is the pre-factor (multiplies from the left) and B the post-factor (multiplies from the right). Pre-multiplying B by A gives AB; post-multiplying gives BA.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 17
What is the condition for the product AB to be defined?

Model Answer
The number of columns of A must equal the number of rows of B (conformable for multiplication). Ex: $A_{3\times4}$, $B_{4\times2}$: AB exists but BA does not.

Grade - Low
Practice map, main concept: Sheet: J-I 11 | DPP: 1(16)
Also used in: Sheet: J-II 5

Question 18
What is the order of the resultant matrix when we multiply two matrices?

Model Answer

$$A_{m\times n}\,B_{n\times p} = (AB)_{m\times p}$$

Ex: orders $1\times3,\ 3\times3,\ 3\times1$ give ABC of order 1×1.

Grade - Low
Practice map, main concept: Sheet: J-II 5 | DPP: 2(2, 15, 18)
Also used in: Sheet: J-I 11 | DPP: 2(16)

Question 19
How do you multiply two matrices?

Model Answer
$(AB)_{ij} = \sum_k a_{ik}b_{kj}$: multiply row i of A with column j of B element-wise and add. Multiplication is associative, $A(BC) = (AB)C$, and distributive, $A(B+C) = AB + AC$.
Ex: $\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \begin{pmatrix}2 & 1\\ 4 & 3\end{pmatrix}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 4-6, 10; J-II 2, 29; C-I 12; C-II 18-19 | DPP: 2(6, 9, 11, 16); 3(1-2, 6); 5(13, 21)
Also used in: Sheet: J-I 3, 8; J-II 12-13, 36-37; J-M 31; C-I 5-7; C-II 7 | DPP: 2(1, 3-5, 8, 10, 12-14, 17) | Question Bank: 155, 158, 164

Question 20
Does the order of appearance of matrices in a product matter?

Model Answer
Yes. In general $AB \neq BA$ (and BA may not even be defined). Never swap factors; also $AB = O$ does not imply $A = O$ or $B = O$.
Ex: $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} = \begin{pmatrix}2 & 1\\4 & 3\end{pmatrix}$ but the reverse product is $\begin{pmatrix}3 & 4\\1 & 2\end{pmatrix}$.

Grade - Low
Practice map, main concept: DPP: 2(3, 8)
Also used in: Sheet: J-II 2 | DPP: 3(1)

Question 21
What happens when we multiply any matrix by I or O?

Model Answer

$$AI = IA = A, \qquad AO = OA = O \quad \text{(orders conformable)}$$

Ex: I is the multiplicative identity, O the multiplicative zero; $A^0 = I$ for a square A.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 22
What happens to the determinant value when we multiply two matrices?

Model Answer

$$|AB| = |A|\,|B|, \qquad |A^n| = |A|^n \quad \text{(square matrices of the same order)}$$

Ex: $|P^{-1}AP| = |A|$; $|A^2| = 25 \Rightarrow |A| = \pm 5$; a square of a real matrix always has $|A^2| \geq 0$.

Grade - High
Practice map, main concept: Sheet: J-I 9; J-M 35, 38; J-A 1, 10, 17; C-I 10 | Question Bank: 122
Also used in: Sheet: J-I 24; J-II 16, 34; J-M 39, 43, 45; J-A 22-23; C-I 4, 12 | Question Bank: 159

Question 23
What do you mean by commutation of two matrices?

Model Answer
A and B commute if $AB = BA$ (both square of the same order). A commutes with A^n, A^{-1}, kI, I and any polynomial in A.
Ex: $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}$ commutes with $\begin{pmatrix}a & b\\c & d\end{pmatrix}$ iff $b = \frac{2}{3}c$ and $d = a + c$ (from $AB = BA$ element by element).

Grade - Medium
Practice map, main concept: Sheet: J-I 8; C-II 15
Also used in: Sheet: J-I 7; J-A 5; C-II 31

Question 24

What do you mean by anti-commutation of two matrices?

Model Answer

A and B anti-commute if $AB = -BA$. Then $(A+B)^2 = A^2 + B^2$.

Ex: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: $AB = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -BA$.

Grade - Low

Practice map, main concept: DPP: 2(10, 12)

Also used in: none

Question 25

When can we use the standard algebraic identities of squares and cubes for matrices?

Model Answer

Only when the matrices commute, $AB = BA$: then $(A \pm B)^2 = A^2 \pm 2AB + B^2$, $A^2 - B^2 = (A-B)(A+B)$, $(A+B)^n$ by the binomial theorem. Always valid: $(A+I)^n$ and $(A+kI)^n$.

Ex: $A^2 - B^2 = (A-B)(A+B) \iff AB = BA$; in general $(A+B)^2 = A^2 + AB + BA + B^2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 7 | DPP: 5(12)

Also used in: Sheet: J-I 33; J-A 5; C-I 20; C-II 31

Question 26

When can we define a higher integral power of a matrix?

Model Answer

Only for a square matrix: $A^2 = AA$, $A^m A^n = A^{m+n}$, $(A^m)^n = A^{mn}$, $A^0 = I$. A polynomial $f(A) = a_0 A^n + \cdots + a_n I$ uses I for the constant term.

Ex: $f(x) = x^2 - 5x + 6$, $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$: $f(A) = A^2 - 5A + 6I = \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3; J-M 31; C-I 22 | DPP: 2(13-14); 5(18) | Question Bank: 164

Also used in: Sheet: J-I 4; J-II 6, 10; J-A 9; C-I 14, 18; C-II 2 | DPP: 5(4, 12)

Question 27

How do you handle trigonometric (rotation-type) matrices: product, powers, inverse?

Model Answer

For $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$: $R(\alpha)R(\beta) = R(\alpha+\beta)$, $R(\theta)^n = R(n\theta)$, $R(\theta)^{-1} = R(-\theta) = R(\theta)^T$, $|R| = 1$. The 3×3 form $F(x) = \begin{pmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{pmatrix}$ obeys the same rules: $F(x)F(y) = F(x+y)$, $F(x)^{-1} = F(-x)$.

Ex: $R\left(\frac{\pi}{6}\right)^{12} = R(2\pi) = I$; if the signs are changed to $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, it is $R(-\theta)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 12; J-II 21; J-M 32; C-I 14 | DPP: 5(2, 5, 16) | Question Bank: 118

Also used in: Sheet: J-I 13, 27 | DPP: 2(6, 9) | Question Bank: 117, 126, 130

Characteristic Equation

Question 28

Can you find the characteristic equation of a square matrix of order 2 and 3?

Model Answer

$|A - \lambda I| = 0$. Order 2: $\lambda^2 - (\operatorname{tr} A)\lambda + |A| = 0$. Order 3: $\lambda^3 - (\operatorname{tr} A)\lambda^2 + C\lambda - |A| = 0$, where $C =$ sum of the three principal 2×2 minors. Sum of roots $= \operatorname{tr} A$, product $= |A|$.

Ex: roots $-1, 3$ of $|A - xI| = 0 \Rightarrow \operatorname{tr} A = 2$, $|A| = -3$, $\operatorname{tr}(A^2) = (-1)^2 + 3^2 = 10$.

Grade - Medium

Practice map, main concept: Question Bank: 132

Also used in: Sheet: J-II 4, 19; J-M 48

Question 29

Do you know that a square matrix always satisfies its characteristic equation?

Model Answer

Yes (Cayley–Hamilton): if $p(\lambda) = |A - \lambda I|$ then $p(A) = O$, with the constant term multiplied by I.

Ex: $A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ satisfies $A^3 - 4A^2 + 4A - I = O$ (here $\operatorname{tr} A = 4$, $|A| = 1$).

Grade - Medium

Practice map, main concept: Sheet: J-II 4; J-M 48; J-A 24; C-II 5

Also used in: Sheet: J-A 15 | DPP: 5(15) | Question Bank: 128

Question 30

Can you write the characteristic equation of a square matrix of order 2 directly?

Model Answer

$$A^2 - (\operatorname{tr} A)\, A + |A|\, I = O$$

Ex: $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$: $\operatorname{tr} A = 5$, $|A| = -2 \Rightarrow A^2 - 5A - 2I = O$.

Grade - High

Practice map, main concept: Sheet: J-M 40, 51; J-A 15, 21 | DPP: 2(1); 5(14-15)

Also used in: Sheet: J-I 19-20, 36; J-M 8, 17, 21, 52-53; J-A 24; C-II 5, 14 | Question Bank: 140, 145

Question 31

Are you able to use the characteristic equation to compute higher powers and in long-division questions?

Model Answer

Replace A^2 by $pA - qI$ repeatedly to lower the power. For a polynomial: divide $f(x)$ by the characteristic polynomial $c(x)$, $f(x) = c(x)Q(x) + R(x)$, then $f(A) = R(A)$.

Ex: $A^2 = 5A + 2I \Rightarrow A^3 = 5A^2 + 2A = 27A + 10I$.

Grade - Medium

Practice map, main concept: Sheet: J-M 8, 18 | Question Bank: 140, 146

Also used in: Sheet: C-II 5 | Question Bank: 156

Question 32

How do you find A^n, an entry, the trace or the sum of all elements of A^n (or of $A + A^2 + \cdots + A^n$)?

Model Answer

(a) Compute A^2, A^3 and spot the pattern (prove by induction). (b) $A = \lambda I + N$ with $N^k = O$: expand binomially. (c) $A^2 = pA + qI$: reduce powers. (d) $A = PDP^{-1} \Rightarrow A^n = PD^nP^{-1}$; for orthogonal P, $Q = PAP^T$ gives $Q^n = PA^nP^T$. (e) Rank-one $A = uv^T$: $A^n = (v^Tu)^{n-1}A$. Sum of all elements $= \mathbf{1}^T A\mathbf{1}$; if every row sums to 1, then $A^n\mathbf{1} = \mathbf{1}$.

Ex: $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: $A^2 = 2A$, so $A^n = 2^{n-1}A$ and the sum of all elements of A^n is 2^{n+1}; a 3×3 matrix with every row sum 1 has A^3 with total sum 3.

Grade - High

Practice map, main concept: Sheet: J-M 6, 12, 16, 20, 26, 53; J-A 25; C-I 1-2, 18; C-II 2, 29 | DPP: 5(1) | Question Bank: 120, 124, 126, 128, 133, 138, 144, 150, 156

Also used in: Sheet: J-M 44 | Question Bank: 116, 118

Important Definition

Question 33

What do you mean by an idempotent matrix and its application in the expansion of $(I + A)^n$?

Model Answer

$A^2 = A$. Then $A^n = A$ for $n \geq 2$, $|A| = 0$ or 1 and $(I + A)^n = I + (2^n - 1)A$. Also $I - A$ is idempotent and $A(I - A) = O$.

Ex: $(I + A)^n = I + 127A \Rightarrow 2^n - 1 = 127 \Rightarrow n = 7$. If $AB = B$, $BA = A$ then $A^2 = ABA = A$, $B^2 = B$, so $A^2 + B^2 = A + B$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2; J-II 7, 11, 37; C-I 17 | DPP: 5(8)

Also used in: Sheet: J-II 8; J-M 12 | Question Bank: 144

Question 34

What do you mean by a periodic matrix and its application in the expansion of $(I + A)^n$?

Model Answer

$A^{k+1} = A$ for some positive integer k; the least such k is the period (an idempotent matrix has period 1). Powers then repeat: $A^{k+r} = A^r$, so in $(I + A)^n = \sum \binom{n}{r} A^r$ group the terms by power and reduce.

Ex: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has $A^3 = A$, so its period is 2.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 16

Question 35

What do you mean by an involutory matrix and its application in the expansion of $(I + A)^n$?

Model Answer

$A^2 = I$ (so $A^{-1} = A$, $|A| = \pm 1$). Then $A^{2m} = I$, $A^{2m+1} = A$ and $(I + A)^n = 2^{n-1}(I + A)$. Also $\frac{1}{2}(I \pm A)$ are idempotent.

Ex: for a real 2×2 involutory $A \neq \pm I$: $\operatorname{tr} A = 0$ and $|A| = -1$ (from $A^2 - (\operatorname{tr} A)A + |A|I = I$).

Grade - Medium

Practice map, main concept: Sheet: J-I 22, 36; J-II 8, 10; J-M 17 | DPP: 4(13) | Question Bank: 145, 155

Also used in: Sheet: J-I 23, 28-29

Question 36
What do you mean by a nilpotent matrix and its application in the expansion of $(I+A)^n$?

Model Answer
$A^m = O$, $A^{m-1} \neq O$; m is the index ($\leq$ order). Every triangular matrix with zero diagonal is nilpotent. Then $(I+A)^n = I + nA + \binom{n}{2}A^2 + \dots$ stops at A^{m-1}. Write $A = \lambda I + N$ and expand.

Ex: $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = I + N$, $N^2 = O$, so $B^{50} = I + 50N = \begin{pmatrix} 1 & 50 \\ 0 & 1 \end{pmatrix}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-II 6, 9; J-A 9 | Question Bank: 116
Also used in: Sheet: J-M 12, 44; J-A 25; C-I 1 | Question Bank: 144

Transpose of a Matrix

Question 37
How do you take the transpose of a matrix?

Model Answer
Interchange rows and columns: $(A^T)_{ij} = a_{ji}$; the order $m \times n$ becomes $n \times m$.

Ex: $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 38
How do you denote the transpose of a matrix?

Model Answer
A^T or A'. Basic rules: $(A^T)^T = A$, $(kA)^T = kA^T$, $(A \pm B)^T = A^T \pm B^T$, $|A^T| = |A|$, $\operatorname{tr} A^T = \operatorname{tr} A$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 2(18)

Question 39
Do you know the pattern when we take the transpose of two or more matrices in a product?

Model Answer

$$(AB)^T = B^T A^T, \qquad (ABC)^T = C^T B^T A^T, \qquad (A^n)^T = (A^T)^n$$

Ex: A, B symmetric $\Rightarrow (AB - BA)^T = BA - AB = -(AB - BA)$, so $AB - BA$ is skew-symmetric.

Grade - **High**
Practice map, main concept: Sheet: J-M 11, 24; J-A 2, 4; C-I 5, 7, 15 | DPP: 2(17, 19); 3(3-4); 5(22) | Question Bank: 143, 148
Also used in: Sheet: J-I 11; J-II 5, 38; J-M 7, 38, 42-43; J-A 7; C-II 29 | DPP: 3(2, 8) | Question Bank: 122, 129, 139

Symmetric and Skew Symmetric Matrix

Question 40

How do you determine whether a square matrix is symmetric?

Model Answer

$A^T = A$. For any square A: $A + A^T$, AA^T, A^TA are symmetric; A^n and A^{-1} of a symmetric matrix are symmetric; AB is symmetric for symmetric A, B iff $AB = BA$.

Ex: $\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$ is symmetric; $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$ is symmetric.

Grade - Medium

Practice map, main concept: Sheet: J-II 12 | DPP: 3(7-8)

Also used in: Sheet: J-I 23; J-M 7, 11, 24; J-A 2, 4, 6-7; C-I 15 | DPP: 3(5); 4(6); 5(22) | Question Bank: 139, 143, 148

Question 41

How do you determine whether a square matrix is skew-symmetric?

Model Answer

$A^T = -A$. For any square A: $A - A^T$ is skew-symmetric. A^{odd} of a skew matrix is skew, A^{even} is symmetric; $AB - BA$ is skew for symmetric A, B.

Ex: $\begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$ is skew-symmetric; its square $\begin{pmatrix} -9 & 0 \\ 0 & -9 \end{pmatrix}$ is symmetric.

Grade - Medium

Practice map, main concept: Sheet: J-A 7

Also used in: Sheet: J-M 7; J-A 4; C-I 15; C-II 3 | DPP: 3(7-8); 5(22) | Question Bank: 139

Question 42

Do you know the determinant value of a skew-symmetric matrix of odd order?

Model Answer

It is 0: $|A| = |A^T| = |-A| = (-1)^n|A| = -|A|$ for odd n. (For even order the determinant need not be 0.)

Ex: $\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix} = 0.$

Grade - Low

Practice map, main concept: DPP: 5(7)

Also used in: none

Question 43

Do you know the condition on the elements of a symmetric matrix?

Model Answer

$a_{ij} = a_{ji}$ for all i, j; only $\frac{n(n+1)}{2}$ elements are free, so with s allowed values there are $s^{n(n+1)/2}$ symmetric matrices.

Ex: 3×3 symmetric matrices with entries from $\{0, 1, \ldots, 9\}$: 10^6.

Grade - Medium

Practice map, main concept: Sheet: J-M 27 | Question Bank: 151

Also used in: Sheet: J-II 12-13, 15; J-M 13, 52; J-A 6, 29; C-II 20, 23 | DPP: 3(5) | Question Bank: 131, 161

Question 44
Do you know the condition on the elements of a skew-symmetric matrix?

Model Answer
$a_{ij} = -a_{ji}$, so every diagonal element is 0 and the trace is 0; only $\frac{n(n-1)}{2}$ elements are free. A matrix that is both symmetric and skew-symmetric is O.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 13; J-A 29; C-II 23

Question 45
How do you express any square matrix as the sum of a symmetric and a skew-symmetric matrix?

Model Answer

$$A = \tfrac{1}{2}(A + A^T) + \tfrac{1}{2}(A - A^T) \quad \text{(symmetric + skew; unique)}$$

Ex: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{pmatrix} + \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$.

Grade - Low
Practice map, main concept: Sheet: J-II 14
Also used in: DPP: 3(3)

Question 46
What do you do in a question when a matrix is given as symmetric or skew-symmetric?

Model Answer
Use $A^T = \pm A$: equate $a_{ij} = \pm a_{ji}$ (skew: diagonal 0, tr $= 0$, $|A| = 0$ if odd order); transpose the given relation; test the parity of products with $(AB)^T = B^T A^T$ and of powers with $(A^n)^T = (A^T)^n$. Ex: if $P^T = 2P + I$ then also $P = 2P^T + I$; substituting gives $P = 4P + 3I$, so $P = -I$.

Grade - Medium
Practice map, main concept: Sheet: J-II 13, 15; J-M 7, 13; C-II 31 | DPP: 3(5) | Question Bank: 129, 139, 161
Also used in: Sheet: J-M 11 | Question Bank: 143

Orthogonal Matrix

Question 47
Are you able to define an orthogonal matrix?

Model Answer
$AA^T = A^T A = I$, i.e. $A^{-1} = A^T$. Then $|A| = \pm 1$; A^T, A^{-1} and products of orthogonal matrices are orthogonal.

Ex: $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ has $AA^T = I$.

Grade - Medium
Practice map, main concept: Sheet: J-I 13-14; J-II 16, 38; J-M 33; C-II 3, 11 | DPP: 5(6) | Question Bank: 121
Also used in: Sheet: J-I 15, 29; J-M 6, 20, 42; C-I 2; C-II 13, 29 | DPP: 3(2, 6); 5(5) | Question Bank: 138, 158

Question 48
What do the elements of the rows/columns of an orthogonal matrix represent?

Model Answer
Rows (and columns) are unit vectors, mutually perpendicular: $\sum_j a_{ij}^2 = 1$ and $\sum_j a_{ij}a_{kj} = 0$ for $i \neq k$.

Ex: $\frac{1}{3}\begin{pmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ 2 & -2 & 1\end{pmatrix}$ is orthogonal; if $AA^T = 9I$ for $\begin{pmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{pmatrix}$ then $(a, b) = (-2, -1)$.

Grade - Low
Practice map, main concept: Sheet: J-I 15
Also used in: none

Question 49
Do you know the relation of the trace of AA^T with the elements of A?

Model Answer

$$\operatorname{tr}(AA^T) = \operatorname{tr}(A^TA) = \sum_{i,j} a_{ij}^2 \quad \text{(sum of squares of all elements)}$$

Ex: for real A, $\operatorname{tr}(AA^T) = 0 \Rightarrow A = O$; $\operatorname{tr}(M^TM) = 5$ means the squares of the nine entries of a 3×3 matrix M add up to 5.

Grade - Medium
Practice map, main concept: Question Bank: 158
Also used in: Sheet: J-A 11 | Question Bank: 131, 160

Adjoint of a square Matrix

Question 50
How do you find the adjoint of a square matrix?

Model Answer
$\operatorname{adj} A = [C_{ij}]^T$: the transpose of the matrix of cofactors $C_{ij} = (-1)^{i+j}M_{ij}$.

Ex: $\operatorname{adj}\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 5 & 6 & 0\end{pmatrix} = \begin{pmatrix}-24 & 18 & 5\\ 20 & -15 & -4\\ -5 & 4 & 1\end{pmatrix}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 46 | DPP: 4(4)
Also used in: none

Question 51
Do you know a fast method to determine the adjoint of a square matrix of order 2?

Model Answer

$$\operatorname{adj}\begin{pmatrix}a & b\\ c & d\end{pmatrix} = \begin{pmatrix}d & -b\\ -c & a\end{pmatrix} \quad \text{(swap the diagonal, change signs of the other two)}$$

Ex: $\operatorname{adj}\begin{pmatrix}4 & -1\\ 2 & 3\end{pmatrix} = \begin{pmatrix}3 & 1\\ -2 & 4\end{pmatrix}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 18-19; J-M 10, 44 | Question Bank: 142
Also used in: Sheet: J-M 28 | DPP: 3(10); 4(7) | Question Bank: 152

Question 52

Basic theorem $A\,\mathrm{adj}\,A = |A|I = \mathrm{adj}\,A \cdot A$ and its use in proving properties of the adjoint.

Model Answer

Take determinants: $|A||\,\mathrm{adj}\,A| = |A|^n \Rightarrow |\,\mathrm{adj}\,A| = |A|^{n-1}$. If $|A| \neq 0$: $A^{-1} = \frac{\mathrm{adj}\,A}{|A|}$ and $\mathrm{adj}\,A = |A|A^{-1}$. If $|A| = 0$: $A\,\mathrm{adj}\,A = O$.

Ex: $A = \begin{pmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{pmatrix}$: $A\,\mathrm{adj}\,A = |A|I = I$, so $k = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 17; J-A 16 | DPP: 3(9); 4(1-2)

Also used in: Sheet: J-I 18, 25; J-M 46; J-A 3, 8, 20 | Question Bank: 154

Question 53

Are you able to write all 6 properties of the adjoint and prove them?

Model Answer

For A of order n: (i) $|\,\mathrm{adj}\,A| = |A|^{n-1}$; (ii) $\mathrm{adj}(\mathrm{adj}\,A) = |A|^{n-2}A$; (iii) $|\,\mathrm{adj}(\mathrm{adj}\,A)| = |A|^{(n-1)^2}$; (iv) $\mathrm{adj}(AB) = \mathrm{adj}\,B\,\mathrm{adj}\,A$, $\mathrm{adj}(A^m) = (\mathrm{adj}\,A)^m$; (v) $\mathrm{adj}(kA) = k^{n-1}\,\mathrm{adj}\,A$; (vi) $\mathrm{adj}(A^T) = (\mathrm{adj}\,A)^T$.

Ex: 3×3, $|A| = 2$: $|\,\mathrm{adj}(\mathrm{adj}\,A)| = 2^4 = 16$ and $|\,\mathrm{adj}(2A)| = |4\,\mathrm{adj}\,A| = 4^3 \cdot 2^2 = 256$.

Grade - High

Practice map, main concept: Sheet: J-I 16, 28, 31; J-II 17-18; J-M 3, 5, 14, 19, 22-23, 28-30, 36, 41, 43, 45, 47, 54; J-A 3, 20; C-II 1, 16 | DPP: 3(10); 4(3, 5-6); 5(20) | Question Bank: 125, 137, 147, 152-153, 157, 162

Also used in: Sheet: J-M 10; J-A 4, 16; C-II 30 | DPP: 4(11) | Question Bank: 142

Inverse of a Square Matrix

Question 54

What do you mean by the inverse of a square matrix?

Model Answer

If a square B exists with $AB = BA = I$, then B is the (unique) inverse of A. A one-sided relation $AB = I$ for square matrices already gives $BA = I$.

Ex: $A^2 = I \Rightarrow A^{-1} = A$; $A^2 = A^{-1} \Rightarrow A^3 = I$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23; J-A 23; C-I 4

Also used in: Sheet: J-M 1, 49; C-I 22 | DPP: 4(13) | Question Bank: 134

Question 55

How do you denote the inverse of a square matrix A?

Model Answer

A^{-1} (not $\frac{1}{A}$). It satisfies $AA^{-1} = A^{-1}A = I$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 56
What is the condition for a square matrix to be invertible?

Model Answer
$|A| \neq 0$ (non-singular). Then A^{-1} is unique. If $|A| = 1$ and all entries are integers, then A^{-1} also has integer entries.

Ex: $\begin{pmatrix} 2 & 4 \\ 1 & \lambda \end{pmatrix}$ is non-invertible only for $\lambda = 2$.

Grade - Low

Practice map, main concept: DPP: 4(8-9)

Also used in: Sheet: J-I 35; C-I 9

Question 57
How do you find the inverse of a square matrix with the help of the adjoint?

Model Answer

$$A^{-1} = \frac{\operatorname{adj} A}{|A|} \quad (|A| \neq 0)$$

Ex: $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$: $|A| = 1$, $A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$. For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 21, 26-27, 30, 32, 35; J-A 8 | DPP: 4(7, 10, 12); 5(19) | Question Bank: 130, 154

Also used in: Sheet: J-I 12; J-II 20-21, 23-25, 34-35; J-M 55; C-I 21; C-II 4, 7, 10 | DPP: 5(9, 16)

Question 58
What does the inverse of the inverse of A mean?

Model Answer

$$(A^{-1})^{-1} = A$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 59
What is the determinant value of the inverse of A?

Model Answer

$$|A^{-1}| = \frac{1}{|A|} = |A|^{-1}$$

Ex: $|A| = 2$ (order 3): $|\operatorname{adj} A| = 4$, $|A^{-1} \operatorname{adj} A| = \frac{1}{2} \cdot 4 = 2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 24 | DPP: 5(10)

Also used in: Sheet: J-I 25; J-M 14 | Question Bank: 157, 162

Question 60
What do you mean by negative integral powers of A?

Model Answer

$$A^{-n} = (A^{-1})^n = (A^n)^{-1}, \qquad A^m A^{-n} = A^{m-n}$$

Ex: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: $A^{-2} = (A^{-1})^2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$.

Grade - Low

Practice map, main concept: Sheet: J-II 20

Also used in: DPP: 4(11)

Question 61
How do you take the inverse of A^T?

Model Answer

$$(A^T)^{-1} = (A^{-1})^T$$

Grade - Low

Practice map, main concept: DPP: 4(11)

Also used in: Sheet: J-I 28

Question 62
Do you know the pattern when we take the inverse of two or more matrices in a product?

Model Answer

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (ABC)^{-1} = C^{-1}B^{-1}A^{-1}, \qquad (kA)^{-1} = \tfrac{1}{k}A^{-1}$$

Ex: $|P^{-1}AP - 2I| = |P^{-1}(A - 2I)P| = |A - 2I|$; $(A + B)^{-1} \neq A^{-1} + B^{-1}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 25; J-M 49 | Question Bank: 159

Also used in: Sheet: J-I 22; J-M 6, 35, 45; J-A 17, 23; C-II 3, 11 | Question Bank: 120, 130, 138

Question 63
When can we cancel a matrix on both sides of an equation and what does this actually mean in terms of matrix multiplication?

Model Answer

$AB = AC \Rightarrow B = C$ only if A is invertible, and it means pre-multiplying both sides by A^{-1} (same side!); $BA = CA \Rightarrow B = C$ means post-multiplying by A^{-1}. For singular A cancellation fails.

Ex: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: $A \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = A \cdot O = O$ but $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \neq O$.

Grade - Medium

Practice map, main concept: Sheet: J-M 1 | Question Bank: 134

Also used in: Sheet: C-II 6

Question 64

Can you write both points of the property $AB = O$, prove them, and use them in questions?

Model Answer

If $AB = O$: (1) if A is non-singular then $B = O$ (multiply by A^{-1}), and if B is non-singular then $A = O$; (2) hence if both $A \neq O$ and $B \neq O$, both are singular, $|A| = |B| = 0$. Use it as (factor)(factor) $= O$ in tricks such as $P^3 = Q^3$, $P^2Q = Q^2P \Rightarrow (P^2 + Q^2)(P - Q) = O$.

Ex: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = O$ with both factors singular; for 3×3 non-zero A, B with $AB = O$ the system $AX = O$ has infinitely many solutions.

Grade - Medium

Practice map, main concept: Sheet: J-I 33; J-II 39; J-A 5; C-I 6, 20; C-II 8, 14 | DPP: 2(7); 5(11)

Also used in: Sheet: C-II 6

Question 65

How do you use the characteristic equation to find the inverse of a square matrix?

Model Answer

From $A^2 - pA + qI = O$ with $q = |A| \neq 0$, multiply by A^{-1}: $A^{-1} = \frac{1}{q}(pI - A)$. In general solve $p(A) = O$ for I and factor out A. The same trick turns $3A^3 + 2A^2 + 5A + I = O$ into $A^{-1} = -(3A^2 + 2A + 5I)$.

Ex: $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $A^2 - 5A - 2I = O \Rightarrow A^{-1} = \frac{1}{2}(A - 5I) = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 20; J-II 19; J-M 21, 52 | DPP: 5(17)

Also used in: Sheet: J-A 15

Finding inverse by elementary operations

Question 66

How do you find the inverse by elementary row operations?

Model Answer

Write $[A \,|\, I]$ and apply only row operations ($R_i \leftrightarrow R_j$, $R_i \to kR_i$, $R_i \to R_i + kR_j$) until the left block becomes I; the right block is then A^{-1}. A row of zeros on the left means A is not invertible.

Ex: $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$: $R_2 \to R_2 - 2R_1$, $R_2 \to -R_2$, $R_1 \to R_1 - 2R_2$ give $A^{-1} = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 67

How do you find the inverse by elementary column operations?

Model Answer

Write A above I and apply only column operations ($C_i \leftrightarrow C_j$, $C_i \to kC_i$, $C_i \to C_i + kC_j$) until the upper block is I; the lower block is A^{-1}. Never mix row and column operations.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 68
What should you keep in mind while using row operations on both sides of a matrix equation?

Model Answer
Write $A = IA$ and apply each row operation to the left side and to the pre-factor I of the right side (never to A on the right): $A = IA \to I = BA$, so $B = A^{-1}$. Do not use column operations in the same calculation.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 69
What should you keep in mind while using column operations on both sides of a matrix equation?

Model Answer
Write $A = AI$ and apply each column operation to the left side and to the post-factor I of the right side: $A = AI \to I = AB$, so $B = A^{-1}$. Do not use row operations in the same calculation.

Grade - Low
Practice map, main concept: none
Also used in: none

System of Equations

Question 70
Can you write a given system of equations as $AX = B$ and vice versa?

Model Answer
A = coefficient matrix, X = column of unknowns, B = column of constants. If $|A| \neq 0$: $X = A^{-1}B$ (unique solution). Homogeneous $AX = O$ has a non-zero solution iff $|A| = 0$.

Ex: $x + 2y = 5,\ 3x + 4y = 11 \iff \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$, giving $x = 1,\ y = 2$.

Grade - Medium
Practice map, main concept: Sheet: J-I 37; J-II 24-27; J-M 55; C-II 7, 26-28 | Question Bank: 123
Also used in: Sheet: J-I 34, 38; J-M 37; J-A 12-13, 26; C-II 4, 9, 12, 21-22 | Question Bank: 127, 129, 133

Question 71
Are you able to relate $AX = B$ with Cramer's rule?

Model Answer
$|A| \neq 0$: $X = \frac{(\text{adj}\, A)B}{|A|}$, i.e. $x_i = \frac{\Delta_i}{\Delta}$ (Δ_i: column i of $|A|$ replaced by B). $|A| = 0$: if $(\text{adj}\, A)B \neq O$ the system is inconsistent; if $(\text{adj}\, A)B = O$ it has infinitely many solutions or none (check further).
Ex: $x + y + z = 6,\ x - y + z = 2,\ 2x + y - z = 1$: $\Delta = 6$, $\Delta_1 = 6$, so $x = 1$; similarly $y = 2,\ z = 3$.

Grade - Medium
Practice map, main concept: Sheet: J-A 12-13; C-II 9, 12
Also used in: Sheet: J-I 37; J-II 24; C-II 26-28

Question 72

How do you use a matrix defined by its action on column matrices ($AX_1 = Y_1,\ AX_2 = Y_2, \dots$)?

Model Answer

Put the given columns side by side: $A[X_1\ X_2\ X_3] = [Y_1\ Y_2\ Y_3] \Rightarrow A = YX^{-1}$ (if X invertible); if $X_i = e_i$ the columns of A are the Y_i. If $AX = \lambda X$ then $A^nX = \lambda^n X$ and $(A - kI)X = (\lambda - k)X$: use it to solve or classify $(A - kI)Y = B$.

Ex: $A\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}2\\3\end{pmatrix}$ and $A\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix} \Rightarrow A = \begin{pmatrix}2 & 4\\3 & 5\end{pmatrix}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 34; J-II 22, 34-36; J-M 37, 39, 42, 50 | Question Bank: 127

Also used in: none

Question 73

How do you solve a matrix equation such as $AX = B$, $XA = B$ or $AXB = C$ for the unknown matrix X?

Model Answer

$AX = B \Rightarrow X = A^{-1}B$ (pre-multiply); $XA = B \Rightarrow X = BA^{-1}$ (post-multiply); $AXB = C \Rightarrow X = A^{-1}CB^{-1}$. If a factor is singular, write X with unknown entries and compare elements: the solution may not exist or may not be unique.

Ex: $A = \begin{pmatrix}2 & 1\\1 & 1\end{pmatrix}$, $B = \begin{pmatrix}1 & 0\\2 & 3\end{pmatrix}$: $AX = B \Rightarrow X = A^{-1}B = \begin{pmatrix}-1 & -3\\3 & 6\end{pmatrix}$.

Grade - Low

Practice map, main concept: Sheet: J-II 23; C-I 21; C-II 4, 6, 10

Also used in: none

Rank of Matrix

Question 74

What does the rank of a matrix mean?

Model Answer

The order of the highest-order non-singular square sub-matrix. If rank is r: every sub-matrix of order $r + 1$ or more is singular and at least one of order r is non-singular; rank $\leq \min(m, n)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 75

How do you find the rank of a matrix?

Model Answer

Row-reduce to echelon form and count the non-zero rows, or find the largest non-vanishing minor: $|A| \neq 0 \Rightarrow$ rank n; otherwise test the 2×2 minors, then the non-zero entries.

Ex: $\begin{pmatrix}2 & -1 & 3\\1 & 4 & 0\\3 & 3 & 3\end{pmatrix}$: $R_3 \to R_3 - R_1 - R_2$ gives a zero row, so rank $= 2$.

Grade - Medium

Practice map, main concept: Sheet: J-II 31-33

Also used in: Sheet: J-II 26-27; J-A 13; C-II 9, 12

Extra Reading Material

Question 76

What is a permutation matrix? Write down its properties.

Model Answer

A matrix obtained by permuting the rows of I: each row and each column has exactly one 1, the rest 0. Properties: $P^TP = I$ (orthogonal, $P^{-1} = P^T$), $|P| = \pm 1$ (sign of the permutation), product of permutation matrices is a permutation matrix, $\operatorname{tr} P =$ number of fixed positions; there are $n!$ of order n.

Ex: 5×5 matrices of $0, 1$ with exactly one 1 in every row and every column are the permutation matrices: $5! = 120$.

Grade - Medium

Practice map, main concept: Sheet: J-I 29; J-M 4; J-A 18, 22 | Question Bank: 136

Also used in: none

Question 77

What is the effect on a matrix if we pre-multiply or post-multiply it by a permutation matrix?

Model Answer

PA permutes the rows of A; AP permutes the columns of A (by the same pattern as the rows of I were permuted).

Ex: $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: PA swaps the two rows of A, AP swaps the two columns.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 18, 22

Question 78

Which power of a permutation matrix is I?

Model Answer

$P^k = I$ where k is the order of the permutation, the lcm of its cycle lengths (so $P^{n!} = I$ always).

Ex: the cyclic matrix $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ has $P^3 = I$, so $P^{49} = P$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 79

Are you able to write X^TAX directly, where X is a column matrix and A a square matrix?

Model Answer

$X^TAX = \sum_{i,j} a_{ij}x_ix_j$ is a 1×1 matrix (a quadratic form). For symmetric A: $\sum a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j$; for skew-symmetric A it is 0.

Ex: $\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + 4xy + 3y^2$.

Grade - Low

Practice map, main concept: DPP: 2(4)

Also used in: none

Miscellaneous (added items)

Question 80

How do you count (or find the extreme value of) matrices whose entries come from a given set and satisfy a condition?

Model Answer

Turn the condition into conditions on the entries and count cell by cell. Useful translations: $\operatorname{tr}(A^TA) = \sum a_{ij}^2$; symmetric $n \times n$ with s values $\Rightarrow s^{n(n+1)/2}$; $|A| \neq 0$ (count the complement $|A| = 0$); sum of entries = prime (list the possible sums). For the maximum of $|A|$ use bounds (Hadamard / AM-GM) and try a candidate.

Ex: 2×2 matrices with entries in $\{0, 1, 2\}$: 50 of the 81 are invertible (so $P = \frac{50}{81}$); for entries in $\{0, 1\}$ exactly 6 of 16 are invertible.

Grade - High

Practice map, main concept: Sheet: J-I 38; J-M 15, 25; J-A 11, 14, 27-29; C-I 19; C-II 13, 20-25, 30 | Question Bank: 131, 149, 160, 163

Also used in: Sheet: J-M 4, 27, 50 | Question Bank: 121, 136, 151